INCH-POUND
MIL-DTL-17547B(SH)
18 September 2018
SUPERSEDING
MIL-V-17547A(SH)
9 November 1988

DETAIL SPECIFICATION

VALVES, CHECK, BRONZE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes bronze check valves of various types for low pressure air, water, and oil services.

1.2 Classification. Check valves are of the following types and classes as specified (see 6.1 and 6.2):

1.2.1 Types.

a. Type A – Swing
b. Type B – Lift horizontal
c. Type C – Lift vertical
d. Type D – Lift angle
e. Type E – Tilting disc (top entry)
f. Type F – Tilting disc (in line)
g. Type G – Split disc (wafer type)
h. Type H – Split disc (lug type)

1.2.2 Classes.

a. Class 1 – For working pressure 50 pounds per square inch (lb/in^2) max, 14 inches and larger (for types E, F, G, and H).
b. Class 2 – For working pressure 250 lb/in^2 max, 2½ to 12 inches (for types A, B, C, D, G, and H).
c. Class 3 – For working pressure 400 lb/in^2 max, ¼ to 2 inches (for types A, B, C, and D).
d. Class 4 – For working pressure 400 lb/in^2 max, 2½ to 10 inches (for types A, B, C, and D).

1.3 Part or identifying number (PIN).

M	**17547**	**-**	**X**	**X**	**XXX**	**/**	**XXX**	**XX**
Prefix for Military Specification	Specification Number	-	Type	Class	Rating	/	Nominal Size	Potable water use

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

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Type		Class		Rating (psi)		Nominal size (inches)		Potable water use	
PIN Code	Type	PIN Code	Composition	PIN Code	Rating	PIN Code	Size [1/]	PIN Code	Category
A	Swing	1	Class 1	005	50	002	¼	PW	Suitable
B	Lift horizontal	2	Class 2	015	150	003	⅜		Not suitable
C	Lift vertical	3	Class 3	030	300	005	½		
D	Lift angle	4	Class 4	040	400	007	¾		
E	Tilting disc (top entry)					010	1		
F	Tilting disc (in line)					012	1¼		
G	Split disc (wafer type)					020	2		
H	Split disc (lug type)					025	2½		
						030	3		
						040	4		
						Size 6-46 [2/]			
						Code = “nps0”	See table III		

NOTES:

[1/] For additional information regarding nominal size for types, see table III.

[2/] For sizes 6-8, code shall show “0(size)0”, and for sizes 10-46, code shall show “(size)0”.

Example: M17547-C1030/010PW is for a vertical lift check valve, class 1, 300 psi, 1 inch, suitable for potable water use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

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FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-59491 - Sealing Compound, Thread and Gasket, Hydrocarbon Fluid and Water Resistant

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-PRF-1149 - Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness

MIL-DTL-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plates, Identification

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-L-24131 - Lubricant, Colloidal Graphite in Isopropanol

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.GSAglobalsupply.gsa.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385637 - Valves, Bronze, Flanged, Check 250 PSI WOG

803-1385721 - Valves, Bronze, 1/4-2, Union End, Swing Check

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

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NAVSEA PUBLICATIONS

0900-LP-001-7000	-	Fabrication and Inspection of Brazed Piping Systems
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection and Casting and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/ AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://webstore.ansi.org/.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.24	-	Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves: Classes 150, 300, 600, 900, 1500, and 2500
ASME B18.2.1	-	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
ASME B18.2.2	-	Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B98/B98M	-	Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes
ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B164	-	Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B271/B271M	-	Standard Specification for Copper-Base Alloy Centrifugal Castings
ASTM B443	-	Standard Specification for Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625) and Nickel-Chromium-Molybdenum-Silicon Alloy (UNS N06219) Plate, Sheet, and Strip

(Copies of these documents are available online at www.astm.org.)

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NSF INTERNATIONAL

NSF/ANSI 61 - Drinking Water System Components - Health Effects

(Copies of this document are available online at www.nsf.org.)

SAE INTERNATIONAL

SAE HS1086 - Numbering Metals and Alloys

SAE AMS3216 - Elastomer: Fluorocarbon (FKM) Rubber, High-Temperature - Fluid Resistant Low Compression Set 70 to 80

SAE AMS4676 - Nickel-Copper Alloy, Corrosion-Resistant, Bars and Forgings 66.5Ni - 3.0Al - 0.62Ti - 28Cu Hot-Finished, Precipitation Hardenable

SAE AMS4892 - Alloy Castings, Corrosion and Heat-Resistant 66Ni - 29Cu - 4.0Si As Cast

SAE AMS-C-6183 - Cork and Rubber Composition Sheet; for Aromatic Fuel and Oil Resistant Gaskets

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless otherwise specified (see 6.2), valve materials shall be in accordance with table I. However, this specification is not intended to be restrictive provided proposed alternative materials will give equal or better service than the material specified. Proposed alternative materials shall be subject to approval by NAVSEA.

TABLE I. List of materials.

Valve parts	Material	Specification or standard	Remarks
Body Cap or cover	Valve bronze	ASTM B61 (UNS C92200) for sizes under 16 inches only	1/
	Nickel-aluminum-bronze	ASTM B148 alloy 958 (UNS C95800), ASTM B271/B271M, all sizes	
Seat Hinge pin Disc nut Disc nut pin Disc washer Lever plug Hanger or lever Disc	Nickel-copper alloy	QQ-N-281, ASTM B164 (UNS N04400), QQ-N-286, SAE AMS 4676 (UNS N05500), ASTM A494/A494M grade M25S, SAE AMS 4892 (UNS N04019)	1/, 2/, 3/
	Valve bronze	ASTM B61 (UNS C65100)	
Threaded fasteners	Silicon bronze	ASTM B98/B98M (UNS C65100)	ASME B18.2.1 ASME B18.2.2 1/
	Nickel-copper alloy	QQ-N-281, ASTM B164 (UNS N04400)	

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TABLE I. List of materials – Continued.

Valve parts	Material	Specification or standard	Remarks
Gasket	Nitrile and cork sheet	SAE AMS-C-6183, class 1, grade C firm	For JP-5
	Fluorocarbon	SAE AMS 3216	For oil and air service
	Synthetic rubber	MIL-PRF-1149, type I, class 2	For water service

NOTES:

1/ This table contains the commercial and Military specifications and standards, and, in addition, contains the corresponding SAE HS1086 Unified Numbering System (UNS) designation. Reference to a specific UNS designation will permit the use of any one of the many alloy specifications and standards listed in SAE HS1086 for that UNS number.

2/ Valves for seawater service shall be provided with nickel-copper alloy trim, such as discs, seats, disc nuts, hinge pins, side plugs, hanger or lever, and washers as applicable. The hinge pins shall be made from QQ-N-286 material and the lever/hanger shall be made from QQ-N-281 or ASTM A494/A494M materiel.

3/ For valves where low permeability (non-magnetic) material is required, QQ-N-286 shall be used for valve trim.

3.2.1 Prohibited materials. The following materials shall be prohibited: magnesium and its alloys, cadmium plating, asbestos, and any radioactive materials. The use of cadmium-plated parts and fasteners, including washers, is also prohibited. Lead and its compounds shall not be used where they can come in contact with seawater feed, permeate, or freshwater. The lead content in bronze alloys shall not exceed levels allowed by ASTM B61. All materials that may come in contact with potable freshwater shall be NSF/ANSI 61 approved for drinking water contact.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Service rating. Maximum gauge pressure and maximum temperature for valves shall be as specified in table II.

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TABLE II. Service rating.

Type	Class	Nominal pipe size (nps) (inches)	Maximum gauge pressure (lb/in^2)	Maximum temperature (°F)
A	2	2½ to 12 inclusive; 250 lb/in^2 maximum	See 803-1385637	
	3	¼ to 2 inclusive; 400 lb/in^2 maximum	See 803-1385721	
	4	2½ to 10 inclusive; 400 lb/in^2 maximum	400 360 320 280 240 200 150	150 1/ 200 250 300 350 400 425
B, C, and D	2	2½ to 12 inclusive; 250 lb/in^2 maximum	250 235 215 200 180 160 150	150 1/ 200 250 300 350 400 425
	3	¼ to 2 inclusive; 400 lb/in^2 maximum	400 360 320 280 240 200 150	150 1/ 200 250 300 350 400 425
	4	2½ to 10 inclusive; 400 lb/in^2 maximum		
E, F, G, and H	1	14 and larger 50 lb/in^2 maximum	50	150 1/

NOTE:

1/ The service pressure rating at temperatures less than 150 °F shall be the same as the rating shown for 150 °F. Valves used at these lower temperatures shall conform to the requirements applicable to the condition under which it is used.

3.4 Construction.

3.4.1 Body walls. The walls of valve bodies shall not have flat surfaces. Ribbing shall be provided as necessary to prevent distortion.

3.4.1.1 Threads. Threads on threaded parts shall comply with FED-STD-H28.

3.4.2 Discs. Discs shall be either of cast or forged nickel-copper alloy or bronze, with an insert of nickel-copper alloy (cast, forged, or sheet), in accordance with table I, securely silver brazed to the disc to form the seating surface. Silver brazing shall be in accordance with 0900-LP-001-7000. The finished thickness of the insert shall be at least ⅛ inch for valves 2½ to 6 inches, 5⁄32 inch for valves 7 to 12 inches, 3⁄16 inch for valves 14 to 22 inches, ¼ inch for valves 24 to 30 inches, and ⅜ inch for valves 32 to 40 inches.

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3.4.3 Seat. Segmented seat rings shall not be permitted. All valves should have sufficient seat material to initially lap the seats. When possible, valves should have sufficient material for relapping as expected over the service life of the valve.

3.4.4 Body. Valve bodies shall be one-piece construction.

3.4.5 End connections. Unless otherwise specified (see 6.2), valves in sizes 2 inches and below shall be furnished with silver-brazing union ends in accordance with MIL-DTL-1183. Valves in sizes 2½ inches and larger shall be furnished with flanged ends. The flanges shall be cast integrally with the valve body, and the inlet and outlet flange shall be of the same size and pressure rating.

3.4.6 Flanges. Flanges shall be in accordance with MIL-PRF-20042, using corresponding pressure ratings. Unless otherwise specified (see 6.2), valves shall be supplied with flanges drilled in accordance with MIL-PRF-20042, except where an odd number of bolt holes are specified (see 6.2). The flange bolt holes shall be equally spaced and positioned such that the full section cutting plane bisects the angle between the holes. Flanges shall be spot-faced or finished on the back, parallel to the flange face within 1 degree for the bolt heads and nuts. The spot face shall not cut more than 1/32 inch below the minimum plate thickness. Eccentricity of the spot face diameter with the bolt hole shall be minimized to provide facing for the bolt head. Diameter for spot facing shall be 110 to 120 percent of the bolt head corner-to-corner dimension. Rims need not be finished, provided they do not have surface discontinuities such as cracks, sand inclusions, gas holes, or excessive shrinkage.

3.4.7 Face-to-face dimension. Flanged valves shall have face-to-face and face-to-end dimensions as specified in table III.

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TABLE III. Dimensions and weights of flanges valves.

Valve size (nps) (inches)	Types A, B, C face-to-face (inches)	Type D face-to-end (inches)	Types A, B, C, D max weight (pounds)	Type E		Type F		Types G & H face-to-face (inches)	Type G max weight (pounds)	Type H max weight (pounds)
				Face-to-face (inches)	Max weight (pounds)	Face-to-face (inches)	Max weight (pounds)			
2½	10	6¼	48	---	---	---	---	2⅝	11	19
3	10¾	6¾	58	---	---	---	---	2⅞	15	38
4	12¼	7¾	78	---	---	---	---	2⅞	18	48
5	13¾	8¾	110	---	---	---	---	3⅜	35	80
6	15¼	9½	150	---	---	---	---	3⅞	45	96
8	18½	11½	275	---	---	---	---	5	82	174
10	21¾	13¼	330	---	---	---	---	5¾	125	238
12	25¼	14¾	500	---	---	---	---	7⅛	200	330
14	---	---	---	20	662	12¾	464	8¾	325	422
16	---	---	---	20¾	850	13	595	9⅛	415	537
18	---	---	---	21¾	1,030	13¼	721	10⅜	555	648
20	---	---	---	26	1,232	13¾	862	11½	725	893
22	---	---	---	26½	1,430	14	1,001	---	---	---
24	---	---	---	28	1,633	14¼	1,143	12½	1,100	1,500
26	---	---	---	30	1,730	14½	1,211	---	---	---
28	---	---	---	31	1,850	16	1,295	---	---	---
30	---	---	---	34	1,923	16¼	1,346	---	---	---
32	---	---	---	34¾	2,150	16½	1,505	---	---	---
34	---	---	---	36¼	2,350	17½	1,645	---	---	---
36	---	---	---	38	2,598	18	1,818	---	---	---
38	---	---	---	39¾	2,900	19	2,030	---	---	---
40	---	---	---	40¾	3,130	20	2,191	---	---	---
42 1/	---	---	---	41½	3,430	22	2,403	---	---	---
46	---	---	---	44½	3,855	24¼	2,800	---	---	---

NOTE:

1/ Sizes 42 inches and larger shall be as approved by NAVSEA.

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3.4.8 Type A, class 4 valves (swing check). Type A, class 4 (2½ to 12 inches, 400 lb/in^2 maximum) valves shall permit an unobstructed flow and the area at any point within the valve in the line of flow shall be equal to the inside diameter of the standard nps. The valves shall operate satisfactorily when installed in either the vertical or horizontal position.

3.4.9 Types B, C, and D valves (lift check). Lift check valves shall be of a design with body configuration as specified (see 1.2).

3.4.10 Types E and F valves (tilting disc check). Types E and F valves shall be as specified in 3.4.10.1 through 3.4.10.3.

3.4.10.1 Dashpot device. The valves shall incorporate a dashpot device, which will provide a cushioned closure for minimum slam and maximum tightness.

3.4.10.2 Hinge pin. Valves shall be designed for installation in a horizontal run of pipe such that the hinge pin is above the horizontal.

3.4.10.3 Type E valves. Type E valves shall be of an elliptical or rectangular top-entry with a bonnet cap that will allow replacement of the disc and seat without the need for removal of the valve body from the piping system.

3.4.11 Types G and H valves (split disc check).

3.4.11.1 Plate springs. Plate springs for seawater applications shall be in accordance with ASTM B443.

3.4.11.2 Flanges. Types G and H valves shall be constructed to be assembled with ASME B16.24 flanges.

3.4.11.3 Lugs, type H valves. For type H valves, lugs shall be either threaded or clear drilled as specified (see 6.2).

3.4.11.4 Thread sealant. Thread sealant for threaded parts, except where silver brazing alloy is applied, shall be sealing compound in accordance with A-A-59491. A very thin coat of sealing compound shall be applied to threads and bearing surfaces. The sealant compound may also be used to seal gasket at body and cap joint. No other gasket sealing compound is permitted.

3.4.11.5 Lubrication. Lubrication of valve parts shall be with graphite in isopropanol in accordance with MIL-L-24131.

3.4.12. Welding. Unless otherwise specified (see 6.2), weld repair and inspection of nickel-aluminum-bronze castings shall be in accordance with S9074-AR-GIB-010/278.

3.5 Shock resistance. Valves shall withstand high impact shock in accordance with grade A, hull-bulkhead mounted principal unit, complete assembly, class I, type A of MIL-DTL-901.

3.6 Hydrostatic pressure. Valves shall withstand hydrostatic pressure of 150 percent of the valve rated pressure (at 150 °F) as specified in table II.

3.7 Seat leakage. Leakage rate through a closed check valve seat shall not exceed 25 cubic centimeters (cm^3) per hour per inch of nps size. When specified (see 6.2), where nominal leakage cannot be tolerated, seat leakage shall not exceed 10 cm^3 per hour per inch of nps.

3.8 Vibration resistance. The valves shall withstand vibration in accordance with type I of MIL-STD-167-1.

3.9 Body marking and identification plates.

3.9.1 Body marking. The following markings shall be integrally cast on the valve body:

a. Manufacturer’s name or trademark

b. Valve size in inches

c. Valve pressure rating in lb/in^2

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d. MIL-DTL-17547

e. Direction of flow arrow

3.9.2 Identification plate. Valves shall be provided with an identification plate in accordance with MIL-DTL-15024 and MIL-P-15024/5, types A, B, C, or D. Identification plates made of corrosion-resisting steel or brass shall be permanently fastened to a part of the valve not subjected to working pressure. The identification plate shall contain the following:

a. Manufacturer's name or trademark

b. Type, class, and size of valve in inches and rating in lb/in^2

c. Body and material composition

d. Valve trim material (disc, seat, and so forth)

e. Manufacturer's identification number (Commercial and Government Entity [CAGE])

f. MIL-DTL-17547

g. National Stock Number (NSN)

3.9.2.1 Order of precedence. Identification plates may be reduced in size for the smaller valves by limiting the contents to items (b) and (c). Identification plates, when attached, shall not cover the integrally cast marking.

3.10 Flanged valves dimensions, weights, and flow coefficient. Dimensions, weights, and flow coefficient (C_v) shall be as specified in 3.10.1 through 3.10.2.

3.10.1 Dimensions and weights. Dimensions and weights for valves 2½ inches and larger shall be as specified in table III.

3.10.2 Flow coefficient. The flow C_v for valves shall be as specified in table IV. The flow rates are based on values for valves in the fully open position.

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TABLE IV. Minimum C_v.

Valve size (nps)	Type A swing check, classes 2, 3, and 4	Types B, C, and D lift check, class 2, 3, and 4			Types E and F tilting disk check, class 1	Types G and H
		Horizontal	Vertical	Angle		
¼	4	1	3	2	---	---
⅜	7	2	6	3	---	---
½	12	3	11	6	---	---
¾	20	6	19	10	---	---
1	35	10	33	17	---	---
1¼	61	17	58	30	---	---
1½	82	23	78	41	---	---
2	139	40	133	69	---	---
2½	214	61	204	107	---	82.1
3	316	91	302	158	---	139.3
4	543	156	518	271	---	271.3
5	877	253	836	438	---	482.5
6	1,287	371	1,227	643	---	725
8	2,255	651	2,150	1,127	---	1,509
10	3,566	1,029	3,400	1,783	---	2,640
12	5,200	1,501	4,958	2,600	---	4,075
14	---	---	---	---	7,200	5,303
16	---	---	---	---	10,000	7,340
18	---	---	---	---	12,300	9,940
20	---	---	---	---	16,000	12,960
22	---	---	---	---	19,500	---
24	---	---	---	---	23,500	20,500
26	---	---	---	---	27,000	---
28	---	---	---	---	32,000	---
30	---	---	---	---	38,000	---
32	---	---	---	---	43,500	---
34	---	---	---	---	49,750	---
36	---	---	---	---	56,250	---
38	---	---	---	---	62,400	---
40	---	---	---	---	68,000	---
42	---	---	---	---	69,000	---
46	---	---	---	---	80,874	---

3.11 Maintainability. Valve design and construction shall assure that assembly and disassembly of the valve can be accomplished without the need for special training or special tools. Special tools are defined as those tools not listed in the GSA Global Supply Catalog. Design and external configuration shall be such as to permit the use of a portable boring machine to repair the bonnet inlay of the valve with the valve in place in the piping system.

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3.12 Workmanship. All parts shall be free of adhering sand, hard spots, excessive shrinkage, cracks, or scale. Unacceptable visual surface discontinuities such as cracks, tears, laps, sand inclusions, and gas holes shall be removed and their removal shall be verified by visual examination (see 4.5.2). Castings shall not be repaired, plugged, impregnated, brazed, burned in, or peened, except as specified in 3.4.12.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. One valve of each type and class shall be subjected to first article inspection. Acceptance of the inspected sizes shall include the sizes not inspected as shown in table V. First article inspection shall consist of the examination and tests specified in table VI. Additionally, every valve offered to the Government shall be tested in accordance with 4.6.4.

TABLE V. Valve test sizes.

Size tested (inches)	Approves these sizes (inches)
¾	¾ and smaller
2	1 to 2, inclusive
6	2½ to 6, inclusive
8	7 and 8
10	9 and 10
over 10	All sizes shall be tested

TABLE VI. Examination and tests.

Inspection	Requirement	First article	Conformance
Materials	3.2 and 3.2.2	4.5.1	4.5.1
Visual and dimensional examination	3.4, 3.9, 3.10, and 3.12	4.5.2	4.5.2
Examination after first article tests	3.11	4.5.3	
Shock	3.5	4.6.1	
Vibration	3.8	4.6.2	
Operation	3.3	4.6.3	4.6.3
Hydrostatic pressure and seat leakage	3.6 and 3.7	4.6.4	4.6.4
Disc silver brazing	3.4.2	4.6.5	
Prohibited materials	3.2.1	4.5.1	

4.3 Conformance inspection. Every valve offered to the Government shall be tested in accordance with 4.6.4. Further conformance inspection shall be performed on samples selected in accordance with 4.4. This inspection shall include the examinations and tests specified in table VI.

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4.4 Sampling.

4.4.1 Sampling for examination and tests. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and Acceptance Quality Limit (AQL) as specified (see 6.2).

4.4.2 Lot. A lot shall consist of all valves of the same type, class, and size produced under the same conditions and offered for delivery at one time, as specified (see 6.2). A sample valve shall consist of one completely assembled check valve.

4.5 Examinations.

4.5.1 Examination of materials. Materials used in the manufacture of valves shall be examined to determine conformance to 3.2 and table I, as applicable (see 6.2).

4.5.2 Visual and dimensional examination. Sample valves selected shall be visually and dimensionally examined to verify compliance with the nonoperational requirements of this specification. Each of the sample valves selected shall be examined in accordance with table VI and table VII. Any valve in the sample containing one or more defects shall be rejected, and if the number of defective valves in any sample exceeds the acceptable number for that sample, the lot represented by the sample shall be rejected.

4.5.3 Examination after first article tests. Upon conclusion of first article tests, the valve shall be disassembled and visually and dimensionally examined for damage and wear. Any valve or sample lot not meeting defect criteria of table VIII shall be rejected unless repairable in accordance with 4.6.6. Disassembly and reassembly shall be performed to determine the practicability of maintaining a valve of this design for shipboard use. A maintainability demonstration shall be conducted by removing and replacing the pressure seal ring, or seal welded flexible seal ring, disc, and spring. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used. Particular emphasis shall be directed toward the possibility of loss of small parts by the maintenance crew.

4.6 Tests.

4.6.1 Shock. The valve shall be subjected to the grade A shock test in accordance with MIL-DTL-901 to determine conformance to 3.5. Valves shall be hydrostatically pressurized internally to the maximum gauge pressure as specified in table II (see 3.6). The testing medium shall be clean tap water at temperatures not less than 40 °F or exceeding 125 °F. The testing medium shall not contain additives except corrosion inhibitors, which do not increase the viscosity or surface tension of the water, and residual cutting oil from machining. Momentary malfunction at the time of impact blow is permitted and acceptable. Permanent deformation, misalignment, or functional impairments shall be cause for rejection. Permissible seat leakage shall be as specified in 3.7.

4.6.2 Vibration. The valve shall be subjected to the type I vibration test in accordance with MIL-STD-167-1 to determine conformance to 3.8.

4.6.3 Operational tests. Operational tests shall be performed on each valve size and type. Operational tests shall be conducted in accordance with the test conditions in 4.6.4.1, alternately applying the pressure specified in 3.6 with the pressure on the seat with the valve in the check position for 2,000 cycles. One cycle shall consist of the above specified pressure applied in the check position and in the open position. At the completion of the operational test, the seat tightness test of 4.6.4.3 shall be performed.

4.6.4 Hydrostatic pressure and seat leakage. Every valve offered to the Government shall be tested in accordance with 4.6.4.1 through 4.6.4.3.

4.6.4.1 Test conditions. Each valve shall be hydrostatically tested for structural integrity and seat leakage. The testing medium shall be clean tap water at temperatures not less than 40 °F or exceeding 125 °F. The testing medium shall not contain additives except corrosion inhibitors, which do not increase the viscosity or surface tension of the water, and residual cutting oil from machining. Test fixtures shall be adjusted to permit valve body expansion and elongation due to internal pressure to the fullest extent possible. The thrust on the ends shall be the minimum required to facilitate testing and prevent leakage to the atmosphere.

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4.6.4.2 Hydrostatic pressure test method. Each valve shall be hydrostatically pressurized internally in the open position to 150 percent of the valve rated pressure specified in table II. Pressure shall be maintained until a thorough visual inspection has been conducted, but not for less than one minute. Any external weeping or leakage during pressurizing or permanent deformation after pressurizing shall be cause for rejection. The appearance of water or wetting at the fluid boundary formed by the gasket shall be cause for rejection.

4.6.4.3 Seat tightness test. Each valve shall be subjected to the test pressure, as specified herein below, with the disc in the closed position. If no leakage occurs after one minute of pressurization, the test may be discontinued. If there is any visible leakage, the test shall be continued for a sufficient length of time to accurately determine the rate of leakage. Leakage shall not exceed that specified in 3.7. **Failure to meet the requirements of 3.7 is cause for rejection.**

Valve primary pressure rating	Test in check position
150 lb/in^2 and below Over 150 lb/in^2	50 lb/in^2 100 lb/in^2

4.6.5 Disc silver brazing. Disc silver brazing shall be inspected in accordance with 0900-LP-001-7000 to determine conformance to 3.4.2.

4.6.6 Repair by welding. Repairs shall be made using procedures and welder qualifications in accordance with 3.4.12. Weld repairs shall be inspected in accordance with S9074-AR-GIB-010/278.

TABLE VII. Classification of defects.

Categories	Major defects
101	Type or class not as specified (see 1.2)
102	Material not as specified (see 3.2)
103	Construction not as specified (see 3.4)
104	Shock resistance not as specified (see 3.5)
105	Hydrostatic withstand pressure not as specified (see 3.6)
106	Seat leakage not as specified (see 3.7)
107	Vibration resistance not as specified (see 3.8)
108	Body marking and identification plates not as specified (see 3.9)
109	Flanged valves dimensions, weights, and flow not as specified (see 3.10)
110	Maintainability not as specified (see 3.11)
111	Workmanship not as specified (see 3.12)

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Types A, B, C, and D valves are intended for naval shipboard service primarily for water, oil, or gas service. Types E and F valves are intended for use on naval shipboard seawater services as main injection and overboard discharge valves.

6.1.1 Existing stock. Current stock of valves may be used until depleted; future requisitions for valves should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size (see 1.2).
c. When first article is required (see 3.1).
d. Material composition if other than specified (see 3.2).
e. If end connection of valve sizes 2 inches and below should be furnished other than specified (see 3.4.5).
f. Flange drilling if other than specified and if odd number of bolt holes is required (see 3.4.6).
g. For type H valves, if lugs are to be threaded or clear drilled (see 3.4.11.3).
h. Welding if other than specified (see 3.4.12).
i. When seat leakage other than nominal is required (see 3.7).
j. AQL of 1.0 (see 4.4.1).
k. Lot size being ordered (see 4.4.2).
l. Requirement for prohibited material verification (see 4.5.1).
m. Packaging requirements (see 5.1).
n. Recommended documentation (see 6.3).

6.3 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any valve(s) under this specification (see 6.2):

a. Engineering drawings
b. Materials listing
c. Installation drawings
d. Technical manual
e. Test reports
f. On-board repair parts list

6.4 Subject term (key word) listing.

Disc, tilting (in-line)
Disc, tilting (top entry)
Lift, angle
Lift, horizontal
Lift, vertical
Swing

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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